CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

NAAC grade 'A+'

ACADEMIC REGULATIONS & SYLLABUS

Faculty of Science
P. D. Patel Institute of Applied Sciences
Department of Chemical Sciences
Master of Science Programme
(Advanced Organic Chemistry)

P. D. Patel Institute of APPLIED SCIENCES

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Science

ACADEMIC REGULATIONS

M. Sc. (Advanced Organic Chemistry) Programme

Charotar University of Science and Technology (CHARUSAT)

CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand

Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in

www.charusat.ac.in

Year – 2024-25

CHARUSAT

FACULTY OF SCIENCE
ACADEMIC REGULATIONS

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following are the academic rules and regulations.

1. *System of Education*

The Semester system of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to do a specified course work in the chosen subject of specialization and also complete a project/dissertation if any. Medium of instruction will be English

2. *Duration of Programme*

Postgraduate programme	(M.Sc.)
Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

The maximum limit can be extended by 1 or 2 semester subject to the approval of university on case to case basis.

3. *Eligibility for admissions*

For the admission to M.Sc., programs in the subject of Biological/Physical/Mathematical/Chemical Sciences a candidate must have obtained a Degree of Bachelor of Science from any recognized University or a Degree recognized as equivalent thereto, with minimum Second Class.

4. *Mode of admissions*

Admission to M.Sc. programme will be based on the performance at graduation.

5. *Programme structure and Credits*

A student admitted to a program should study the course and earn credits specified in the course structure.

6. *Attendance*

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7. Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 50% of the marks for the course; and

7.1.2 Final examination will be conducted by the University for 70% of the marks for the course.

*7.2 **Internal Evaluation***

7.2.1 Internal evaluation will be based on internal tests and several other tools of assessment like, quiz, viva, seminar etc., as prescribed by concerned teacher and decided by the faculty.

*7.3 **Internal Institutional evaluation for practicals***

7.3.1 One internal practical test/viva will be conducted per semester totaling to 50 % internal marks for practicals

7.3.2 In "Continuous evaluation" Students shall be evaluated in a continuous manner for their involvement in the practical, aptitude for learning, completion of practical related assignments, regularity in the practicals and record keeping

7.3.3 The final examination by the University 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.3.4 In order to earn the credit in a course a student has to obtain grade other than FF.

7.4 Performance at Internal & University Examination

7.4.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course.
Details of minimum percentage of marks to be obtained in the examinations are as follows

Minimum marks in Internal Exam per subject	Minimum marks in University Exam per subject
36%	36%

7.5.2. If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

8.1 The internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Grading Scheme:

Percentage	>=80	>=75 and <80	>=70 and <75	>=65 and <70	>=60 and <65	>=55 and <60	>=50 and <55	<50
Grade	**AA**	**AB**	**BB**	**BC**	**CC**	**CD**	**DD**	**FF**
Grade Point	10	9	8	7	6	5	4	0

For all CHARUSAT UG/PG degree/diploma programs, except Pharmacy programmes				**For all Pharmacy programmes**		
Letter Grade	**Grade Point**	**Grading Scheme for Mark (In %)**		**Letter Grade**	**Grade Point**	**Grading Scheme for Mark (In %)**
		For all CHARUSAT UG/PG degree/diploma programs, except Pharmacy and Nursing programmes	**B.Sc. Nursing, P.B. B.Sc. Nursing and M.Sc. Nursing**			B. Pharm, & M. Pharm.
O (Outstanding)	10	96.0-100	85.0 – 100	**AA** (Outstanding)	10	90.0 – 100
A+ (Excellent)	9	86.0-95.9	80.0 – 84.9	**AB** (Excellent)	9	80.0 – 89.9
A (Very Good)	8	76.0-85.9	75.0 – 79.9	**BB** (Very Good)	8	70.0 – 79.9
B+ (Good)	7	66.0-75.9	65.0 – 74.9	**BC** (Good)	7	60.0 – 69.9
B (Above Average)	6	56.0- 65.9	60.0 – 64.9	**CC** (Average)	6	50.0 – 59.9
C (Average)	5	46.0 – 55.9	50.0 – 59.9	**FF** (Fail)	0	Below 50.0
P (Pass)	4	36.0 – 45.9	-	**Ab** (Absent)	0	Absent
F (Fail)	0	Below 36.0	Below 50.0	-	-	-
Ab (Absent)	0	Absent	Absent	-	-	-

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Chemical Sciences

Faculty of Science

P. D. Patel Institute of Applied Sciences

MASTER OF SCIENCE

VISION

To become a premier national institute imparting science education integrated with research.

MISSION

To engage in education, research, and spread of science for the benefit of society.

PROGRAM EDUCATIONAL OBJECTIVES

The Postgraduate would be able to

PEO-1: Apply the acquired knowledge in various fields of science to plan and execute tasks.

PEO-2: Have scientific aptitude and competency in solving local and global problems.

PEO-3 Possess an ethical approach while executing research projects and scientific writing.

PEO-4 Ability to follow life-long learning.

PEO-5 Understand the importance of teamwork and possess leadership and entrepreneurship traits.

PROGRAM OUTCOMES

The Post Graduates would be able to

PO1	**Knowledge in sciences:** Possess knowledge and comprehend the various core and allied courses offered under the opted stream of sciences.
PO2	**Planning Abilities:** Demonstrate effective planning abilities, including time management, resource management, delegation skills, organizational skills, teamwork, and interpersonal skills.
PO3	**Problem analysis:** Utilize the principles of scientific enquiry and think analytically, critically with clarity of thought, while solving problems.

PO4	**Modern tool usage:** Apply appropriate methodologies, methods and procedures, resources to conceptualize the matters.
PO5	**Leadership skills:** Realize the importance of teamwork and take initiatives to plan and implement a workflow to fulfill tasks with a compassionate attitude.
PO6	**Professional Identity:** Establish as a professional in the chosen area of the profession with social responsibility.
PO7	**Ethics in Science:** To understand the value of being ethical and implement its principles while carrying out research and publication, tasks at workplace and in life. To respect and adhere to the rules/regulations/treaties/directives/notifications/ laws related to Scientific research and business.
PO8	**Communication:** Communicate effectively and responsibly with the scientific community and with society at large, such as being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.
PO9	**Modern Science and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and help society whenever situations arise.
PO10	**Environment and sustainability:** Understand the causes of environmental pollution and how it can be reduced and remedied using green technology in industries and environment.
PO11	**Life-long learning:** Vigilant about changes in the worldwide professional scenario arising out of technological, political, and economic factors. Recognize the need for and undertake steps necessary to fill the gaps recognized to keep oneself professionally competent. Self- assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

P. D. PATEL INSTITUTE OF APPLIED SCIENCES

DEPARTMENT OF CHEMICAL SCIENCES

Teaching and Examination Scheme

M. Sc. ADVANCED ORGANIC CHEMISTRY

(Effective Year 2024-25)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
P. D. PATEL INSTITUTE OF APPLIED SCIENCES
DEPARTMENT OF CHEMICAL SCIENCES

Degree: M.Sc. (AOC)
Semester I

Course code	Course Title	Teaching Scheme				Examination Scheme				
		Contact hours/week			Credits	TH		PR		Total
		L	T	P		Internal	External	Internal	External	
CHUC501	Selected topics of Quantum Chemistry	3	1		4	18/50	18/50			36/100
CHUC502	Stereochemistry and Organic Reactions	3	1		4	18/50	18/50			36/100
CHUC503	Chemical Dynamics, Electrochemistry, and Group Theory	3	1		4	18/50	18/50			36/100
CHUC504	Separation Techniques	3	1		4	18/50	18/50			36/100
CHUS505	Organic Chemistry Laboratory-I			4	2			09/25	09/25	18/50
CHUS506	Inorganic Chemistry Laboratory-I			4	2			09/25	09/25	18/50
CHUS507	Physical Chemistry Laboratory-I			4	2			09/25	09/25	18/50

*Course chosen by students from the list of courses offered by Humanities and Social Sciences Department, Faculty of Management Studies

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
P. D. PATEL INSTITUTE OF APPLIED SCIENCES
DEPARTMENT OF CHEMICAL SCIENCES

Degree: M.Sc. (AOC)
Semester II

Course code	Course Title	Teaching Scheme				Examination Scheme				
		Contact hours/week			Credits	TH		PR		Total
		L	T	P		Internal	External	Internal	External	
CHUC551	Spectroscopy-I	3	1		4	18/50	18/50			36/100
CHUC552	Name Reactions and Regents in Organic Chemistry	3	1		4	18/50	18/50			36/100
CHUC553	Topics in Physical Chemistry	3	1		4	18/50	18/50			36/100
CHUC554	Coordination Chemistry and Magneto Chemistry	3	1		4	18/50	18/50			36/100
CHUS555	Organic Chemistry Laboratory-II			4	2			09/25	09/25	18/50
CHUS556	Inorganic Chemistry Laboratory-II			4	2			09/25	09/25	18/50
CHUS557	Physical Chemistry Laboratory-II			4	2			09/25	09/25	18/50
CHUD558	Introduction to Polymer Science	2			2			09/25	09/25	18/50

*Course chosen by students from the list of courses offered by the Humanities and Social Sciences Department, Faculty of Management Studies.

***Course chosen by students from the list of University Elective courses.

SYLLABI
(Semester – 1)

Course Code: CHUC501	Course Name: Selected topics of Quantum Chemistry		
Credit: 4	Semester: `1		
Contact hours/week	L	T	P
	3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Fundamentals of Quantum Mechanics**	13
2.	**Particle in a 1D and 3D Box and Tunneling**	11
3.	**Rotational and vibrational motion of a particle**	13
4.	**Hydrogen atom**	12
5.	**Many electron atoms & angular momenta**	11

B. Detailed Syllabus

Unit 1 Fundamental of Quantum Mechanics
Introduction to quantum chemistry, Postulated of Quantum chemistry, Wave function & Schrödinger wave equation, Heisenberg uncertainty principle, Spin and exclusion principle, expectation value, Operator concepts in QM, Hamiltonian operator; angular momentum operator, Linear operators commutation of operators, Commutative property; momentum operator Angular momentum operators and their commutation relations, shift operators and their commutation relations
Unit 2 Particle in a 1D and 3D Box and Tunnelling
Particle in one-dimensional box, three-dimensional box, Numerical of the particle in 3D box, Normalization and orthogonalization, Different problem of orthogonally and normalization, Tunnelling effect, Characteristics of wave functions, Numerical
Unit 3 Rotational and vibrational motion of a particle
Particle on a sphere, Normalization of the wave functions, Rotation of a diatomic molecule and problems, One-dimensional harmonic oscillator, Hermit's differential Equation, Recursion formula for the Hermit's, differential equation normalization, Eigen functions, Different problems of Eigen function, Harmonic oscillator, Polynomials of different degree and problems
Unit 4 Hydrogen atom

Hydrogen atom, Laguerre and associated Laguerre, *Problems of* Laguerre and associated Laguerre, Recursion formula for the radial solution, Solving the Θ equation and φ equation, Variation method and its Application to Hydrogen molecule, Numerical
Unit 5 Many-electron atoms & angular momenta
The wave functions of many-electron systems, the He atom; many-electron atoms, Hartree-self consistent field methods, angular momenta in many electron atoms, communication with Hamiltonian, spin-orbit Interaction, energy states of atoms, and term symbols, Different Problems

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance.

D. Learning Resources

1. Quantum Chemistry: Ira N Levine - PHI Publication.
2. Introductory quantum Chemistry: A K Chandra, TMH Publication.
3. Quantum Chemistry: R K Prasad, New age International Publisher.
4. Quantum Chemistry: John P. Lowe, Academic Press

E. Course Outcomes (COs)

CO1	Understand the fundamentals of quantum chemistry
CO2	To explore applications of quantum chemistry in simple systems
CO3	The solution of the Schrodinger equation for different model systems and their correlation with real systems
CO4	Introduction to new operators such as Hermitian and Hamiltonian and their use in the solution of Hydrogen and Hydrogen like atoms

Course Code: CHUC502	**Course Name: Stereochemistry and Organic Reactions**
Credit: 4	**Semester: 1**
Contact hours/week	**L** / **T** / **P**: 3 / 1 / -

L	T	P
3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Stereochemistry**	14
2.	**Reaction mechanism and methods to determine**	12
3.	**Reactive intermediates in organic chemistry**	14
4.	**Addition, Elimination and Substitution reactions**	14
5.	**Aromaticity**	06

B. Detailed Syllabus

Unit 1 Stereochemistry
Optical and geometrical isomerism, Optical activity, chirality and symmetry, Determination of chirality, Different molecular projections and their interconversions, Absolute and relative configuration, Elements of Chirality, Methods of determining configuration, Cahn-Ingold-Prelog system for R/S nomenclature, Optical purity and Enantiomeric excess, Molecules with two (or more) chiral centres and total possible stereoisomers, Compounds containing chiral atom other than carbon, Stereoisomerism in cyclic structures, Prochirality and prostereoisomerism, Homotopic and heterotopic ligands and faces, Stereospecific and stereo-selective reactions
Unit 2 Reaction mechanism and methods to determine
Types of mechanism and reaction, Transition states and intermediates, Thermodynamic and kinetic requirements for reaction, Baldwin rules for ring closure, thermodynamic and kinetic control, Hammond Postulate, Microscopic Reversibility, Marcus theory, Importance of mechanism and methods of determining mechanisms, Determining reaction mechanism through identification of products, Determining reaction mechanism through identification of intermediates, Determining reaction mechanism through isotopic labeling, Determining reaction mechanism through kinetic isotope effects, Determining reaction mechanism through stereochemical evidences, Determining reaction mechanism through cross-over experiment
Unit 3 Reactive intermediates in organic chemistry

Carbocations (carbonium ions and carbenium ions), Structure, geometry and generation of carbocations, Carbocation stability and fate of carbocations, Reactions involving non-classical carbocations, Structure, geometry and generation of carbanions, Stability and reactions of carbanions, Chiral carbanions and tautomerism in carbanions, Structure and generation of free radicals, Stability and reactions of free radicals, Structure, geometry and generation of carbenes, Reactions of carbenes, Structure, reactivity and generation of nitrenes, Reactions of nitrenes, Structure, reactivity, generation and reactions of arynes
Unit 4 Addition, Elimination and Substitution reactions
Addition reactions- orientation, reactivity and mechanism, Markovnikov and anti-Markovnikov addition reactions, Addition reactions (hydro-halo, hydro-hydroxy, and hydro-alkoxy), Addition reactions (dihydro, dihydroxy, dihalo and ozonolysis), Elimination mechanism and orientation, E1-E2-E1cB mechanistic spectrum, Effect of Substrate Structure, attacking base, leaving group and medium on elimination reaction, Pyrolytic Eliminations, Aliphatic substitution, S_E2, S_E1 and S_Ei Mechanism, S_N1, S_N2, S_Ni mechanism, Neighboring-Group mechanism, Aromatic Substitution, S_NAr, S_N1 Mechanism, Orientation and reactivity in mono-substituted benzene rings, Orientation in benzene rings with more than one substituent, Ipso substitution
Unit 5 Aromaticity
Aromaticity and anti-aromaticity, Huckel's rule of aromaticity, Homo-aromaticity (five-, six-, seven- and eight-membered rings), Other systems containing aromatic sextets, Aromaticity of fused rings and Annulenes, Aromatic systems with electron numbers other than six

C. Pedagogy

The topics will be discussed in interactive class room sessions using classical blackboard teaching/molecular models/power-point presentations/video and animations. Weekly tutorial session will be conducted to develop critical thinking and problem solving approach among students. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be taken as a part of continuous evaluation and suggestions will be given to student in order to improve their performance.

D. Learning Resources

1. M. B. Smith and J. March, March's Advance Organic Chemistry, 6th Ed. (John Wiley)
2. J. Clayden, N. Greeves, and S. Warren, Organic Chemistry 2nd Ed. (Oxford)
3. Organic Reactions, Stereochemistry and Mechanism: P.S. Kalsi (New Age.)
4. Principles of Organic Synthesis: R.O.C Norman & J.M. Coxon (ELBS)
5. Mechanism in Organic Chemistry: Peter Sykes (Orient Longman)
6. Modern Methods of Organic Synthesis: W. Carruthers (Cambridge)
7. Organic Reaction Mechanism: V. K. Ahluwalia and R. K. Parashar (Narosa)

E. Course Outcomes (COs)

CO1	Visualize spatial arrangement of molecules and understand the stereochemistry of organic molecules and prostereogenic elements
CO2	Understand the generation of reaction intermediates and their reaction pathways.
CO3	Realize the involvement of reactive intermediates in different organic transformations
CO4	Understand the concept of aromaticity and structure-activity relationship of organic molecules.
CO5	Acquaint students about the basic concepts of substitution, elimination, and addition reactions with stereochemical outcomes of products.

Course Code: CHUC503	Course Name: Chemical Dynamics and Electrochemistry and Group Theory
Credit: 4	Semester: 1
Contact hours/week	L: 3, T: 1, P: -

L	T	P
3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	Thermodynamics and Properties of Solution	11
2.	Statistical Thermodynamics	11
3.	Chemical Kinetics	14
4.	Group theory in Chemistry	12
5.	Electrochemistry	12

B. Detailed Syllabus

Unit 1 Thermodynamics and Properties of Solution
Introduction and laws of thermodynamics, Heat engines, Nernst Heat Theorem, Properties of the solution, Raoult's Law, Positive and Negative deviation from ideal behavior, Partial molar properties: partial molar free energy, partial molar volume, partial molar heat content their significances, Numerical, Gibbs Duhem Equation, Density method for Determination of PMV, Numerical
Unit 2 Statistical Thermodynamics
Introduction of statistical thermodynamics, Introduction, the weight of configuration, Numerical of weight configuration, Relation between weight configuration and entropy, Relation between weight configuration and entropy and temperature, Derivation of Equation of translational partition function, Numerical of the translational partition function, Derivation of Equation of Rotational partition function, Numerical of Rational partition function, Derivation of Equation of Vibrational partition function
Unit 3 Chemical Kinetics
Chemical Kinetics and its scope, Rate of reaction, factors influencing the rate of a reaction, measurements of reaction rates, Differential and integral rate laws, rate laws, Integration rate law for nth-order reaction half lifetime, Numerical, Equilibrium constants for elementary reactions, the temperature dependence of rate constants, Arrhenius equation, the concept of activation energy, reaction mechanisms, Examples of activation energy and rate laws, Uni-molecular reactions, bi-molecular reactions, tri-molecular, Uni-molecular reactions, bi-molecular reactions, tri-molecular, Opposing reactions, Consecutive reactions, Parallel reactions, Fast reaction and Equation Derivation, Numerical
Unit 4 Group theory in Chemistry

Introduction of group theory Concepts of symmetry in the molecule:- Symmetry elements, Symmetry operations, definitions and theorems in group theory, Examples of groups, subgroups and classes, Molecular Point groups, Classification, notation of point groups, Different examples of point groups, Matrix for clockwise and anticlockwise Rotation., Matric for the plane of reflection, Matrix representation of point groups: product and square rule, inverse rule, matrices for C_{3v}, C_{4v}, Construction of character tables:-rules, reducible and irreducible representations, character of a representation, Properties of irreducible representations, orthogonally theorem, character tables for C_{2v}, C_{3v}, C_{4v}, D_{nh},
Unit 5 Electrochemistry
Introduction to electrochemistry and conductance, Different Conductometric titration and numerical, DHO equation and its Application, Nernst Equation derivation, Numerical of Nernst equation, Concentration cell with transport, Concentration cell without transport, Numerical of concentration cell, Cyclic voltammetry, Numerical, Demonstration of electrochemical work station instruments, Practical Application of electrochemical work station

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance

D. Learning Resources

1. Fundamentals of chemical thermodynamics by M.L. Lakhanpal Publisher: Tata McGraw Hill Publishing Company, New Delhi.
2. Thermodynamics by P.C. Rakshit Publisher: The New Book Stall, Calcutta.
3. Principles of physical chemistry by B. R. Puri, L. R. Sharma and M. S. Pathania Publisher: Vishal Publishing Co., Jalandhar, Delhi
4. Physical chemistry by P. W. Atkins and J. de. Paula Publisher: Oxford Press.
5. Statistical thermodynamics by M. C. Gupta Publisher: Wiley-Eastern Ltd. New Delhi.

E. Course Outcomes (COs)

CO1	***Evaluate the*** *thermodynamic properties the gas and solution*
CO2	***Evaluate*** the macroscopic properties of system though the statistical thermodynamics
CO3	*Evaluate the rate of the chemical reaction*
CO4	**Interpret** the spectra of compound using concept of group theory
CO5	***Solve*** *the problems of electrochemical reaction through electrochemistry*

Course Code: CHUC504	Course Name: Separation Techniques
Credit: 4	**Semester: 1**
Contact hours/week	**L** / **T** / **P**: 3 / 1 / -

L	T	P
3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Fundamentals of separation techniques**	10
2.	**Gas chromatography**	12
3.	**High performance liquid chromatography and Ultra performance liquid chromatography**	14
4.	**Liquid-liquid Partition chromatographic techniques**	14
5.	**Other chromatographic techniques**	10

B. Detailed Syllabus

Unit 1 Fundamentals of separation techniques
Classification of separation techniques, Principle of solvent extraction, Method of separating mixture, Introduction of solvent extraction, Principle of solvent extraction, Distribution coefficient. Advantages and disadvantages of solvent extraction, Solvent extraction techniques**:** Batch extraction, Counter current extraction, Retention time, Retention factor, Height Equivalent to Theoretical plate (HETP), column efficiency, Resolution, Zone broadening, Van deemter's Equation, Discussion of various term in Van deemter's Equation
Unit 2 Gas chromatography
Fundamentals of Gas chromatography, Instrumentation of GC, Stationary phase , carrier gas, Key operation variables controlling retention, Different types of column, Injection port, column component, Detector specification – Flame ionization detector, Thermal conductivity detector, Electron capture detector, Gas density balance detector, Application in the analysis of gases, organic compounds, Determination of Butyl lithium and volatile halogenated hydrocarbons, Determination of carbon in H_2O_2, Isolation and identification of light oil alkanes in shale oil, To determine heat of evaporation of aromatic hydrocarbons, Identification of the components of hetrologous series – Using retention data and functionality test, Determination of empirical formula
Unit 3 High performance liquid chromatography and Ultra performance liquid chromatography

Fundamental of liquid Chromatography separation, Instrumentation, Various Column and guard column, Normal phase and reverse phase chromatography, Isocratic and gradient illusion, Retention factor, Peak Symmetry: Asymmetry Factor and Tailing Factor, Factors affecting column efficiency, UV detector, RI detector, Photo Diode Array detector, ELSD detector, Advantages and disadvantages, Introduction of UPLC, Advantages and disadvantages of UPLC
Unit 4 Liquid-liquid Partition chromatographic techniques
Flash Chromatography: Introduction of flash chromatography, Instrumentation of flash chromatography, Packing of column, selection of solvent, Super critical fluid chromatography: Instrumentation of SFC, stationary and mobile phases used in SFC, Advantages and Application of SFC, Gel permeation chromatography: Instrumentation, Retention behaviour, resolution, Application of GPC, Ion exchange chromatography: Ion exchange resins, structure of resins, Ion exclusion, ligand exchange, Affinity chromatography: Principle and instrumentation, Chromatographic Media, Immobilized Ligand, Advantages and disadvantages of Affinity chromatography
Unit 5 Other chromatographic techniques
Thin Layer Chromatography: Introduction and principle, chromatographic media-coating materials, Activation of adsorbent, Sample development, Development of chromatoplate, solvent systems, Types of development, visualization methods, HPTLC: Introduction, sample application, Micro – preparative chromatography, Quantitative evaluation, Application of HPTLC

C. Pedagogy

The topics will be discussed in interactive class room sessions using classical blackboard teaching to power-point presentations. Special interactive session in the form of seminars will be also conducted by respective faculty members on weekly bases. Each student will give one seminar for a course. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Learning Resources

1. Fundamentals of analytical Chemistry by D. A. Skoog, D. M. West, F. J. Holler, S.R. Crouch, Ninth edition. (CENGAGE Learning)
2. Basic Gas Chromatography by McNiar. (Wiley)
3. HPLC Edited by Yuri Kazakevich & Rosario Lobratto. (Wiley)
4. Separation methods by M N Sastri. (Himalaya Publishing Company)
5. Analytical Chemistry by Gary Christian, seventh edition. (Wiley)
6. Analytical Chemistry: an Introduction by Douglas A. Skoog, Donald M. West and F. James Holler, fifth edition, Saunders college publishing, New York.

7.	Analytical Chromatography by Gurdeep R Chatwal. (Himalaya Publishing Company)
8.	Instrumental method of analysis, H. H. Willard, seventh edition. (Wadsworth publishing company)

E. Course Outcomes (COs)

CO1	Understanding the basic principle of various separation techniques.
CO2	To know about use of analytical techniques like GC, HPLC, HPTLC and other separation techniques in sample analysis.
CO3	To study importance of analytical techniques and its interdisciplinary applications.
CO4	Apply the analytical techniques for qualitative and quantitative analysis of unknown samples

Course Code: CHUS505	Course Name: Organic Chemistry Laboratory I		
Credit: 2	Semester: 1		
Contact hours/week	L	T	P
	-	-	4

A. List of Experiments

Sr. No.	List of Experiments
1.	**Introduction, Record Keeping, and Laboratory Safety** • Introduction • Preparing for the Laboratory • Working in the Laboratory • The Laboratory Notebook • General Protocol for the Laboratory Notebook • Types of Organic Experiments and Notebook Formats • Sample Calculations for Notebook Records • Safe Laboratory Practice: Overview • Safety: General Discussion • Safety: Material Safety Data Sheets (MSDSs) • Safety: Disposal of Chemicals 2
2.	**Techniques and Apparatus** • Glass wares & Assembling Apparatus • Weighing Methods • Filtration Apparatus and Techniques • Solids: Recrystallization and Melting Points • Liquids: Distillation and Boiling Points
3.	**Thin Layer Chromatography** • Introduction & Theory • Determination of R_f Values
4.	**Preparations:** • Dibenzalacetone from Benzaldehyde (Claisen – Schmidt Condensation)

	• Methyl Orange from Sulphanilic acid (Diazotization & Coupling) • 9,10-dihydro anthracene-endo-α,β-succinic anhydride from Anthracene & Maleic anhydride (Diel's Alder Reaction) • p-iodonitrobenzene from p-nitroaniline (Sandmayer Reaction) • P-aminobenzoic acid to p-cholorobenzoic acid (Sandmayer Reaction) • Cyclohexanone to adipic acid (Oxidation) • p-chlorobenzaldehyde to p-chlorobenzoic acid and p-chlorobenzyl alcohol (Cannizzaro Reaction)
5.	**Estimations:** • Determine the amount of Phenol in the given solution. • Determine the amount of Crotonic acid in the given solution.
6.	**Demonstrative Practicals:** • Determination of λ_{max} of the synthesized compound using UV-Visible spectroscopy. • Determination of the presence of functional group using Infra-red Spectroscopy.

B. Pedagogy

The course will give emphasis on active learning in students through utilizing the theoretical knowledge into the practical knowledge. Demonstrative Practicals is helpful to the students to learn the spectroscopic techniques. Students will be continuously evaluated through viva-voce for each experiment and each student will appear in internal test.

C. Learning Resources

1. Elementary practical of Organic Chemistry: A.I. Vogel (Longman group Ltd.)
2. College Practical Chemistry: V K Ahluwalia & Sunita Dhingra (Universities Press (India) Pvt. Ltd.)
3. Practical Organic Chemistry: F. G. Mann & B. C. Saunders (Pearson Education)
4. Experimental Organic Chemistry: John Gilbert & Stephen Martin 6th Ed. (Cengage learning).

5. Advanced Practical Organic Chemistry: J. Leonard, B. Lygo and G. Procter (CRC Press).

D. Course Outcomes (COs)

CO1	Students will learn how to maintain laboratory safety
CO2	They will know the use of different methods for synthesizing the organic compounds.
CO3	They will be prepared for next higher level experiments.

Course Code: CHUS506	Course Name: Inorganic Chemistry Laboratory-I		
Credit: 2	Semester: 1		
Contact hours/week	L	T	P
	-	-	4

A. List of Experiments

1.	Semi-micro qualitative analysis of inorganic mixture (6 + 1 radicals) [max. 6 practical] 6 – Cation / Anion (variable) 1 – Rare earth element from the following Th, Li, Ce, Mo, Te, Se, V, Ti, and Zr
2.	FTIR-sample preparation and data collection (Demonstrative Practical)
3.	Identification of few inorganic compounds through analysis of FTIR spectra

B. Learning Resources

1. Qualitative Inorganic Analysis– A. I. Vogel, 6th Edition revised by G. Svehla ELBS–London
2. Textbook of Chemistry Analysis – A. I. Vogel
3. Qualitative Chemistry semi micro analysis – edited by P. K. Agasyan CBS Publisher-Delhi.
4. Chemistry: Inorganic Qualitative Analysis in the Laboratory, Clyde Metz, Elsevier, 2012.

C. Pedagogy

Hands on experience for the semi-micro qualitative analysis of inorganic mixture through systematic chemical tests on the given mixture. Students will learn sample preparation and data collection using FTIR spectrophotometer. Analysis of FTIR spectra of some inorganic compounds. Internal test and viva will be taken as a part of continuous evaluation.

D. Course Outcomes (COs)

CO1	Understand the fundamentals of qualitative analysis.
CO2	Gain knowledge about the identification of radicals in the inorganic mixture.
CO3	Able to analyze the presence of functional groups through FTIR spectroscopy.
CO4	Acquaint students with semi-micro analysis method.

Course CodeCHUS507	Course Name: Physical Chemistry Laboratory-I
Credit: 2	**Semester: 1**
Contact hours/week	L: - / T: - / P: 4

L	T	P
-	-	4

A. List of Experiments

Sr. No.	Experiments
1.	(1) To find out the cell constant of a given conductivity cell (2) To determine the critical micelle concentration of anionic surfactant. **(Conductometer CMC)**
2.	To verify the additivities of absorbance of a mixture of colored substances in solution using potassium permanganate and potassium dichromate solution**.** (**Spectrophotometer)**
3.	Determination of concentration/ amounts of iodide, bromide and chloride in the mixture by potentiometric titration with silver nitrate. **(Potentiometer)**
4.	Determination of ΔG, ΔH, and ΔS for a reaction using an electrochemical cell. **(Potentiometer)**
5.	To determine the dissociation constants (k1 and k2) for a dibasic acid. **(pH Meter)**
6.	To find the unknown concentration of sugar solution using a polarimeter. (**Polarimeter**)
7.	To determine the rate of acid-catalyst iodination of acetone in the presence of Excess acid & acetone at room temperature. **(Chemical Kinetics)**
8.	To determine the composition of a binary liquid mixture by refractometry. (**Refractometer**)
9.	Determine the surface tension of liquids (methyl acetate, ethyl acetate, Chloroform, hexane ....) and hence calculate the atomic parachors of C, H & Cl. **(Surface Tension)**
10.	To determine Partial Molal Volumes of Sodium Chloride in an aqueous solution at room temperature. **(Partial Molal Volume)**

C. Pedagogy

The students will be taught how to do experiments with instruments like pH meter, conductometer, polarimeter, potentiometer, Refractometer, and spectrophotometer for

exercises in the physical chemistry laboratory. They will also perform non-instrumental experiments to understand the other concept of physical chemistry. Internal tests and viva will be taken as a part of a continuous evaluation

D. Learning Resources

1. Experimental Physical Chemistry by R. C. Das & B. Behera, (Tata McGraw hill Publishing Company Ltd., New Delhi)
2. Systematic Experimental Physical Chemistry by S. W. Rajbhoj and T. K. Chondhekar, (Anjali Publication, Aurangabad)
3. Advanced Practical Physical Chemistry by J. B. Yadav, (Goel Publishing House, Meerut)
4. Experimental Physical Chemistry by G. Peter Matthews, (Clarendon Press, Oxford, London)
5. Experimental Physical Chemistry by V. D. Athawale and ParulMathur, (New Age International Publishers, New Delhi)
6. Advanced Physical Chemistry Experiments by Gurtu and Gurtu, (Pragati Prakashan, Meerut)

E. Course Outcomes (COs)

CO1	Acquire knowledge about different methods of preparation of the solution.
CO2	Students learn how to take measurements with instruments like Refractometer, Conductometer, pH Meter, and Spectrophotometer and interpret the results correctly.
CO3	Understand the concept of chemical Kinetics and thermodynamics through experiments.
CO4	Develop the skill of interpreting data from various experiments, including the construction of appropriate graphs and evaluation of errors.

SYLLABI
(Semester – 2)

Course Code: CHUC551	Course Name: Spectroscopy I
Credit: 4	Semester: 2
Contact hours/week	L: 3, T: 1, P: -

L	T	P
3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	Molecular Formulas and What can be learned from them	05
2.	UV-Visible Spectroscopy	15
3.	FTIR Spectroscopy	15
4.	Mass Spectroscopy	15
5.	Microscopic Techniques	10

B. Detailed Syllabus

Unit 1 Molecular Formulas and What can be learned from them

- Elemental Analysis and Calculations.
- Determination of Molecular Mass.
- Molecular Formulas.
- Index of Hydrogen Deficiency.
- The Rule of Thirteen.

Unit 2 UV-Visible Spectroscopy

- Introduction.
- Laws of Absorption (Lambert-Beer's Law).
- Theory of Electronic transitions and Ultraviolet absorption.
- Chromophores and Auxochromes (Hyper, Hypo, Batho and Hypso Chromic effect).
- Woodward-Fieser rules for Dienes.
- Fieser-Kuhn rule for highly conjugated dienes.
- Woodward-Fieser rules for enones.

- Characteristic absorptions in alkenes and alkynes alcohols, ethers, amines, carbonyl compounds.
- Characteristic absorptions in aromatic compounds.
- λmax Calculation of Aromatic Compounds.
- Factors influencing λmax.
- Effects of conjugation and the effect of solvent.
- Differentiation of compounds / isomers by UV.

Unit 3 FTIR Spectroscopy

- Introduction.
- Theory and principles.
- Molecular vibrations.
- Calculations of vibrational frequencies.
- Factors influencing IR frequency.
- Characteristic group absorptions in hydrocarbons, aromatic compounds, alcohol and phenols, ethers, carbonyl compounds, amines, nitriles, nitro compounds, carboxylic acids and halide.
- Differentiation of compounds/isomers by IR.

Unit 4 Mass Spectroscopy

- Introduction.
- Theory and principles of mass spectroscopy.
- Instrumentation.
- Low and High resolution mass spectra.
- Ionization techniques – Electron Impact (EI) ionization and Chemical Ionization (CI).
- Determination of molecular weight and molecular formula.
- Nitrogen rule.
- Detection of molecular ion peak, metastable ion peak.
- Fragmentations – rules governing the fragmentations.
- Mc Lafferty rearrangement.
- Interpretation of mass spectra of different class of compounds – saturated and unsaturated hydrocarbons, aromatic hydrocarbons, alcohols, ethers,

ketones, aldehydes, carboxylic acids, amines, amides, compounds containing halogens.
Unit 5 Microscopic Techniques • Introduction • Scanning Electron Microscopy (SEM) – Basic Principle, Instrumentation, Operating parameters and Applications • Scanning Tunneling Microscopy (STM) – Basic Principle, Instrumentation, Operating parameters and Applications • Atomic Force Microscopy (AFM) – Basic Principle, Instrumentation, Operating parameters and Applications

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching/molecular models/power-point presentations/videos and animations. The weekly tutorial session will be conducted to develop students' critical thinking and problem-solving approach. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be taken as a part of continuous evaluation, and suggestions will be given to the student to improve their performance.

D. Learning Resources

1. Spectrometric Identification of Organic Compounds: Robert M. Silverstein & Francis X. Webster 7th Ed. (Wiley). **2.** Introduction to Spectroscopy: Donald L. Pavia, Gary M. Lampman, George S. Kriz & James R. Vyvyan 5th Ed. (Cengage Learning). **3.** Spectroscopy of Organic Compounds: P. S. Kalsi 6th Ed. (New Age International Publishers). **4.** Organic Spectroscopy: Willliam Kemp 3rd Ed. (Palgrave Macmillan) **5.** Fundamentals of Molecular Spectroscopy: Colin N. Banwell & Elaine M. McCash 4th Ed. (McGrawHill Education).

6. Principles of Instrumental Analysis: Douglas A. Skoog, F. James Holler & Stanley R. Crouch 7th Ed. (Cengage Learning).

E. Course Outcomes (COs)

CO1	The basic principles involved in the interaction of electromagnetic radiation with matter.
CO2	Gain insight into the basic principles of UV-Visible, FTIR and Mass spectroscopic techniques.
CO3	Understand basic components of UV-Visible, FTIR and Mass Spectrometer.
CO4	Interpretation of UV-Visible, FTIR and Mass Spectra.

Course Code: CHUC552	Course Name: Name Reactions and Reagents in Organic Chemistry			
Credit: 4	**Semester: 2**			
Contact hours/week	**L**	**T**	**P**	
	3	**1**	**-**	

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Organic name reactions, mechanism and application-I**	15
2.	**Organic name reactions, mechanism and application-II**	15
3.	**Reagents in organic synthesis-I**	15
4.	**Reagents in organic synthesis-II**	15

B. Detailed Syllabus

Unit 1 Organic name reactions, mechanism and application-I
Robinson Annulation and related reactions, Importance of Mannich reaction in Robinson Annulation, Stork Enamine reaction, Wittig reaction and its modification, Peterson olefination, Shapiro and Bamford Steven's reaction, Julia Olefination, Vilsmayer-Heck reaction, Mitsonobu reaction, Buchwald-Hartwing amination, Brown's Hydroboration, Hydroboration-oxidation reaction, Applications of hindered boron reagents, Carbonylation of organoboranes, Stobbe condensation
Unit 2 Organic name reactions, mechanism and application-II
Stille coupling, Suzuki coupling, Sonogashira coupling, Heck reaction, Negishi coupling, Mukaiyama esterification, Ritter reaction, Yamaguchi esterification, Domino reactions and its characteristics, Nazerov cyclization, Ugi reaction, Strecker synthesis, Click reactions and its characteristics, Huisgen 1,3-dipolar cycloaddition
Unit 3 Reagents in organic synthesis-I
DCC, LDA, Grignard reagent, Gilman reagent, Alkyl lithium, 1,3-Dithianes, CrO_3, MnO_2, SeO_2, $KMnO_4$, $Pb(OAc)_4$, OsO_4, HIO_4, DMSO, H_2O_2
Unit 4 Reagents in organic synthesis-II
Ozone, HgO, NBS, $K_3Fe(CN)_6$, DDQ, $Al(O\text{-}t\text{-}Bu)_3$, $Al(O\text{-}iPr)_3$, Dess-Martin, $LiAlH_4$, $NaBH_4$, Complex hydrides, DIBAL, Zn-Hg/HCl, $N_2H_4/^{-}OH$, TBTH, Cat. H_2, Wilkinson catalyst

C. Pedagogy

The topics will be discussed in interactive class room sessions using classical blackboard teaching/molecular models/power-point presentations/video and animations. Weekly tutorial session will be conducted to develop critical thinking and problem solving approach among students. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be taken as a part of continuous evaluation and suggestions will be given to student in order to improve their performance.

D. Learning Resources

1. M. B. Smith and J. March, March's Advance Organic Chemistry, 6th Ed. (John Wiley)
2. J. Clayden, N. Greeves, and S. Warren, Organic Chemistry 2nd Ed. (Oxford)
3. Advanced Organic Chemistry: Reaction Mechanisms, R. Bruckner, Academic Press (2002).
4. Principles of Organic Synthesis: R.O.C Norman & J.M. Coxon (ELBS)
5. Organic reactions & their mechanisms, third revised edition, P.S. Kalsi, New Age International Publishers.
6. Modern Methods of Organic Synthesis: W. Carruthers (Cambridge)
7. Organic Reaction Mechanism: V. K. Ahluwalia and R. K. Parashar (Narosa)

E. Course Outcomes (COs)

CO1	Recognize the reaction mechanism and reaction pathways.
CO2	Realize the uses of various types of oxidants and reducing agents in organic synthesis.
CO3	Understand the importance of name reactions and molecular rearrangements.
CO4	Gain knowledge about the industrial applications of name reactions and reagents.
CO5	Acquaint students about the different synthetic strategies.

Course Code: CHUC553	Course Name: Topics in Physical Chemistry		
Credit: 4	Semester: 2		
Contact hours/week	L	T	P
	3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Fugacity and Partial Molar Properties**	13
2.	**Molecular weight determination of Polymers and Kinetics**	13
3.	**Introduction to Nanomaterials**	10
4.	**Surface Tension & Intermolecular Forces**	12
5.	**Solid State Chemistry**	12

B. Detailed Syllabus

Unit 1 Fugacity and Partial Molar Properties
Introduction to Fugacity, Methods for Determination of Fugacity, Relative fugacity, Different Methods for Determination of fugacity for pure gas, Graphical and approximate method, Equation of state method, Numerical for Determination of fugacity, Determination of fugacity of the mixture of gas, Clapeyron - Clausius equation, Partial Molar Properties, Method for Determination of Partial Molar Properties, Numerical
Unit 2 Molecular weight determination of Polymers and Kinetics
Introduction of polymer chemistry, Classification of Mol. Wt. determination methods Based on colligative properties, End group analysis method, Numerical elevation of boiling point and depression in freezing point method, Membrane osmometry and Vapour pressure osmometry method, Derivation of Equation of Numerical, Viscosity, GPC, and light scattering method, Mechanism of Free radical polymerization, Kinetics of free radical polymerization, Kinetics of copolymerization, Reactivity ratio
Unit 3 Introduction to Nanomaterials
Introduction, Fundamentals, Properties and Size dependence of properties, Chemical Optical, vibrational, thermal Electrical Magnetic Mechanical, Nano Material characterization techniques, Applications
Unit 4 Surface Tension & Intermolecular Forces

Introduction of surface phenomena, *Methods for Determination of Surface tension method capillary rise and numerical, Methods for Determination of Surface tension method Drop weight and drop number, numerical,* Surface active agents, Classification of surface-active agents, Micellization, hydrophobic Interaction, Critical micelle concentration (CMC), factors affecting the CMC of surfactants Different methods for Determination of CMC, Determination of surface parameter, Dipole moments & polar molecules Intermolecular forces between molecules, *Numerical*
Unit 5 Solid State Chemistry
Introduction and different cubic structures, Calculation of coordination number, packing fraction and relation between a and r for SC, Calculation of coordination number, Packing fraction and relation between a and r for BCC, Calculation of coordination number, Packing fraction and relation between a and r for FCC, Miller indices and examples, Bragg Equation. X-ray diffraction pattern of the cubic system, Different numerical of Braggs laws, Crystal defects Vacancy defect, Interstitial defect, Impurity defects

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance.

D. Learning Resources

1. Physical Chemistry by P.W. Atkins and J.de Paula, Oxford Press.
2. Principles of Physical Chemistry by B. R. Puri, L.R. Sharma and M. S. Pathania ; Vishal Publishing House.
3. Physical Chemistry by G. M. Barrow, McGraw-Hill
4. Physical Chemistry of Polymers by A. Tagger , Mir Publisher, Moscow ,1980. 39
5. Text Book of Polymer Science by PadmaL. Nayak and S. Lenka, Kalyani Publisher, New Delhi.
6. Text Book of Polymer Science by Fred W. Billmeyer, John Wiley and Sons.
7. The Physics and Chemistry of Nano Solids by Frank J. Owens and Charles P. Poole Jr, Wiley-Interscience, 2008.
8. Introduction to solids by L. V. Azaroff, McGraw-Hill.
9. Solid State Chemistry and its Applications by A. R. West, Wiley 10. Principles of the solid state by H. V. Keer, Wiley –Eastern Ltd.

E. Course Outcomes (COs)

CO1	*Interpret the behaviour of non ideal gas from the fugacity*
CO2	***Evaluate*** the molecular weight of polymer using different techniques
CO3	*Analyze the different properties of nanomaterials*
CO4	Evaluate the surface properties of different compounds
CO5	***Determine the structure and properties of crystalline substance***

Course Code: CHUC554	**Course Name: Coordination chemistry and magneto Chemistry**
Credit: 4	**Semester: 2**
Contact hours/week	

L	T	P
3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Coordination Chemistry: I**	12
2.	**Coordination Chemistry: II**	12
3.	**Bonding in Transition Metal Complexes**	12
4.	**Magnetochemistry I**	12
5.	**Magnetochemistry: II**	12

B. Detailed Syllabus

Unit 1 Coordination Chemistry: I
Fundamentals of coordination, Determination of shape of the d orbitals using different mathematical Equation, Chemistry Concept of crystal field theory (CFT), Ligand field theory (LFT), Splitting of d-orbitals in various octahedral, Tetragonal, elongation and compression in octahedral complexes, John teller distortion, Numerical
Unit 2 Coordination Chemistry: II
Nephelauxetic series, Derivation of terms for closed subshell, Derivation of terms for D, Derivation of terms for f, Configurations, Correlation diagrams for octahedral, tetrahedral system Numerical, Configurations, Correlation diagrams for octahedral
Unit 3 Bonding in Transition Metal Complexes
Tanabe-Sugano diagram for octahedral and tetrahedral stereochemistry of d1 to d9 systems, Determination of Crystal Field Splitting Energy and electronic parameters for octahedral and tetrahedral complexes, Charge Transfer transitions. Molecular orbital energy level diagram for the octahedral and tetrahedral complexes
Unit 4 Magnetochemistry I

Magnetic susceptibility: Sources of paramagnetism, diamagnetic susceptibility, Pascal constants and constitutive corrections Langevin equation, Van Vleck's formula, antiferromagnetism, Types of antiferromagnetism, antiferromagnetic exchange pathways, Ferromagnetism and magnetic domains, molecular field theory of ferromagnetism, magnetic sub lattice, ferrimagnetism and canting.
Unit 5 Magnetochemistry: II
Spin-orbit coupling, Lande's interval rule, quenching of orbital magnetic moment by crystal field, spin-orbit coupling on A and E terms, spin-orbit coupling on T term, Spin paring: spin paring in octahedral complexes; spin paring in non-octahedral complexes; some aspects of spin pairing and cross over the region.

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance.

D. Learning Resources

1. Elements of Magnetochemistry by R. L. Dutta and A. Syamal, Affiliated East-West Press Pvt. Ltd. New Delhi.
2. Introduction to Ligand Fields by B. N. Figgis, Inter science, New York.
3. Physical Methods in Chemistry by R. S. Drago, W. B. Saunders.
4. Inorganic Electronic Spectroscopy by A. B. P. Lever ; Elsevier, Amsterdam.
5. Magnetism and Transition Metal Complexes by F E. Mabbs and D. J. Machin, Chapman and Hall, London.
6. Electronic Absorption Spectroscopy and Related Techniques by D. N. Sathyanarayana; Universities Press, Hyderabad.

E. Course Outcomes (COs)

CO1	Understand the basics of coordination chemistry including coordination numbers and geometry
CO2	Ensuring that students acquire sound knowledge to value the importance of electronic transitions and magnetic property
CO3	Interpret the electronic spectra of coordination compounds explaining color, allowed and forbidden transitions through Orgel and Tanabe-Sugano diagrams
CO4	Demonstrate the basics, spectral and magnetic properties of Lanthanides and Actinides

Course Code: CHUS555	Course Name: Organic Chemistry Laboratory II		
Credit: 2	Semester: 2		
Contact hours/week	L	T	P
	-	-	4

A. List of Experiments

Sr. No.	List of Experiments
1.	**Use of Computer – Chem Draw, Chem-Sketch and ISI-Draw** • Draw the Structure of Simple aliphatic, aromatic and heterocyclic compounds with different substituents. • IUPAC Names • Predict the ^{1}H NMR and ^{13}C NMR data from the software.
2.	**Preparations:** • 2,4-Dihydroxybenzoic acid from Resorcinol (Oxidation) • m-nitroaniline from m-dinitrobenzene (Reduction) • 2,5-dimethylbenzenesulfonic acid. (Sulphonation) • p-nitroanilne from acetanilide (Nitration) (2-step) • Picric acid from Phenol (Nitration) • p-bromonitrobenzene from p-bromo benzene (Nitration) • p-bromoaniline from Acetanilide (Bromination) (2-Step) • 2,4,6-tribromophenol from Phenol (Bromination) • Acetanilide from Acetophenone (Beckman Rearrangement) (2-Step) • Benzilic Acid from Banzoin (Benzil-Benzilic acid rearrangement) (2-Step)
3.	**Estimation:** • Determine the amount of Aniline in the given solution. • To determine percentage purity of Carbonyl compound (Aldehydes & Ketones).
4.	**Demonstrative Practicals:** • Determination of λ_{max} of the synthesized compound using UV-Visible spectroscopy.

	• Determination of the presence of functional group using Infra-red Spectroscopy.

B. Pedagogy

The course will give emphasis on active learning in students through utilizing the theoretical knowledge into the practical knowledge**.** Demonstrative Practicals is helpful to the students to learn the spectroscopic techniques. Students will be continuously evaluated through viva-voce for each experiment and each student will appear in internal test.

C. Learning Resources

1. Elementary practical of Organic Chemistry: A.I. Vogel (Longman group Ltd.) 2. College Practical Chemistry: V K Ahluwalia & Sunita Dhingra (Universities Press (India) Pvt. Ltd.) 3. Practical Organic Chemistry: F. G. Mann & B. C. Saunders (Pearson Education) 4. Experimental Organic Chemistry: John Gilbert & Stephen Martin 6th Ed. (Cengage learning). 5. Advanced Practical Organic Chemistry: J. Leonard, B. Lygo and G. Procter (CRC Press).

D. Course Outcomes (COs)

CO1	Students will learn how to draw and predict the spectroscopic data with help of computer.
CO2	They will know the use of different methods for synthesizing the organic compounds.
CO3	They will be prepared for next higher level experiments.

<table>
<tr><td>Course Code: CHUS556</td><td colspan="3">Course Name: Inorganic Chemistry Laboratory-II</td></tr>
<tr><td>Credit: 2</td><td colspan="3">Semester: 2</td></tr>
<tr><td rowspan="2">Contact hours/week</td><td>L</td><td>T</td><td>P</td></tr>
<tr><td>-</td><td>-</td><td>4</td></tr>
</table>

A. List of Experiments

1.	Synthesis of metal complexes and double salts [Any Four]
	• Hexaminenickel (II) chloride • Tris(thiourea)copper(I) sulphate • Ferrous ammonium sulphate • Ammonium copper(II) sulphate • Tris (acetylacetonato) iron (III) • Cis- bis (glycinato) copper (II) monohydrates • Any other preparation in line with above-mentioned
2.	Quantitative analysis of some synthesized metal complexes and double salts by gravimetric and volumetric methods
3.	Synthesis of nanomaterials [Any Two]
	• Synthesis of metal oxide nanoparticle (Fe_3O_4/ZnO/TiO_2) • Synthesis of polymeric nanocomposite (PbS-PVA/PVP) • Any other preparation in line with above-mentioned
4.	Characterization of synthesized materials by UV-Vis, FTIR, PXRD and TGA (Demonstrative Practical)

B. Learning Resources

1. Mendham, J., Denney, R. C., Barnes, J. D., Thomas, M. and Sivasankar, B. Vogel's Quantitative Chemical Analysis, 6th Edn., (Pearson Education, 2009).

2. Marr, G., Rockett, B. W. Practical Inorganic Chemistry, (Van Nostrand, 1972).

3. Wollins, J. D. Inorganic Experiments, 3rd Edn., (VCH, 1994).

4. Parshall, G. W. (Ed. in Chief). Inorganic Synthesis, Vol. 15, (McGraw Hill, 1974).

C. Pedagogy

Hands on experience for the synthesis of metal complexes, double salts and nanomaterials. Students will learn quantitative analysis by gravimetric and volumetric methods. Training of sophisticated instruments such as UV-Vis, FTIR, PXRD and TGA for the characterization of materials. Internal test and viva will be taken as a part of continuous evaluation.

D. Course Outcomes (COs)

CO1	Understand the chemistry of metal complexes.
CO2	Familiarize with different synthetic approaches to synthesize inorganic materials.
CO3	Able to recognize the right analytical tool for the characterization of inorganic materials.
CO4	Understand the different aspects of quantitative analysis.

Course Code:CHUS557	Course Name: Physical Chemistry Laboratory-II
Credit: 2	Semester: 2
Contact hours/week	L: - T: - P: 4

L	T	P
-	-	4

A. List of Experiments

Sr. No.	Experiments
1.	To determine the molecular composition of ferric – salicylate complex by Job's method. **(Spectrophotometer)**
2.	To determine the cell constant of a given conductance cell and the equivalent conductance of a strong electrolyte (NaCl) at several concentrations and verify the Onsager Equation. **(Conductometer)**
3.	To determine the rate of the saponification of ethyl acetate at different temperatures conductometrically **(Conductometer)**
4.	To measure the Mol. Wt. of Polymer using viscometer. **(Viscosity)**
5.	To verify the law of refraction for a given mixture of glycol + water system. **(Refractometer)**
6.	To determine the transition temperature of Glauber's salt by solubility method. **(Transition Temperature)**
7.	Determination of distribution coefficient of ammonia between Chloroform and water. **(Distribution Coefficient)**
8.	To determine the heat of the solution of the given acid by the solubility method. **(Heat of solution)**
9.	Determination of the critical solution temperature (CST) of the phenol/water system & to study of the effect of additives on CST. **(CST)**
10.	To determine the critical micelle concentration of anionic surfactant using a stalagmometer. **(CMC)**

C. Pedagogy

The students will be taught how to do experiments with instruments like pH meter, conductometer, polarimeter, potentiometer, Refractometer, and spectrophotometer for exercises in the physical chemistry laboratory. They will also perform non-instrumental experiments to understand the other concept of physical chemistry. Internal tests and viva will be taken as a part of continuous evaluation.

D. Learning Resources

1. Experimental Physical Chemistry by R. C. Das & B. Behera, (Tata McGraw hill Publishing Company Ltd., New Delhi)
2. Systematic Experimental Physical Chemistry by S. W. Rajbhoj and T. K. Chondhekar, (Anjali Publication, Aurangabad)
3. Advanced Practical Physical Chemistry by J. B. Yadav, (Goel Publishing House, Meerut)
4. Experimental Physical Chemistry by G. Peter Matthews, (Clarendon Press, Oxford, London)
5. Experimental Physical Chemistry by V. D. Athawale and Parul Mathur, (New Age International Publishers, New Delhi)
6. Advanced Physical Chemistry Experiments by Gurtu and Gurtu, (Pragati Prakashan, Meerut)

E. Course Outcomes (COs)

CO1	They will know the use of many simple instruments to extract valuable information and its Application
CO2	The student will learn about the experiments related to the properties solution and understand its concept.
CO3	Interpret data from various experiments, including the construction of appropriate graphs and evaluation of errors.
CO4	Develop skills in applying the distribution laws for various solutes between different appropriate solvent